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Operating System

9Week [04 November - 10 November]

SOC3010 OS QUIZ4 FALL 2024

Started on	2024-11-15 23:23
State	Finished
Completed on	2024-11-15 23:57
Time taken	33 mins 56 secs
Marks	585.00/600.00
Grade	9.75 out of 10.00 (98%)

Question 1

Correct

Mark 5.00 out of 5.00

The following Resource limit field in process descriptor indicates maximum size of nonswappable memory in bytes.

Select one:

- ☐ 1. RLIMIT_AS
- ☐ 2. RLIMIT_DATA
- ☐ 3. RLIMIT_RSS
- ☒ 4. RLIMIT_MEMLOCK ✓

The correct answer is: RLIMIT_MEMLOCK

Question 2

Correct

Mark 5.00 out of 5.00

Linux uses processes to offer better support for multithreaded applications.

Select one:

- ☐ 1. sleeping
- ☒ 2. lightweight ✓
- ☐ 3. heavyweight
- ☐ 4. waiting

The correct answer is: lightweight

Question 3

Correct

Mark 5.00 out of 5.00

Version 4 of the Internet Protocol (IPv4) defines an IP address as a number.

Select one:

- ☐ 1. 256-bit
- ☐ 2. 64-bit
- ☐ 3. 128-bit
- ☒ 4. 32-bit ✓

The correct answer is: 32-bit

Question **4**

Correct

Mark 5.00 out of 5.00

If any parent process terminates without reaping a child, then the orphaned child will be reaped by

Select one:

- ☐ 1. Zombie process
- ☐ 2. swapper process
- ☐ 3. process scheduler
- ☒ 4. init process ✓

The correct answer is: init process

Question **5**

Correct

Mark 5.00 out of 5.00

..... is a kernel thread created from scratch during the initialization phase of Linux .

Select one:

- ☐ 1. Init process
- ☐ 2. kirq daemon
- ☒ 3. Swapper process ✓
- ☐ 4. schedule

The correct answer is: Swapper process

Question **6**

Correct

Mark 10.00 out of 10.00

In 64-bit architectures, the system administrator can enlarge the maximum PID number up to

Select one:

- ☒ 1. 4,194,303 ✓
- ☐ 2. 32768
- ☐ 3. 4,194,304
- ☐ 4. 65536

The correct answer is: 4,194,303

Question **7**

Correct

Mark 5.00 out of 5.00

A represents a set of sleeping processes, which are woken up by the kernel when some condition becomes true.

Select one:

- ☒ 1. wait queue ✓
- ☐ 2. run queue
- ☐ 3. ready queue
- ☐ 4. stack

The correct answer is: wait queue

Question **8**

Correct

Mark 5.00 out of 5.00

The child process rarely needs to read or modify all the resources inherited from the parent; in many cases, it issues an immediate and wipes out the address space that was so carefully copied.

Select one:

- ☐ 1. clone
- ☐ 2. vfork
- ☐ 3. fork
- ☒ 4. execve ✓

The correct answer is: execve

Question **9**

Correct

Mark 5.00 out of 5.00

Specify the default Port number associated with SQL Server

Answer: 156 ✓

The correct answer is: 156

Question **10**

Correct

Mark 5.00 out of 5.00

The 80x86 microprocessors automatically save the FPU, MMX, and XMM registers in the TSS.

SAY TRUE OR FALSE

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question **11**

Correct

Mark 5.00 out of 5.00

The register of each CPU contains the TSSD Selector of the corresponding TSS.

Select one:

- ☒ 1. tr ✓
- ☐ 2. gdtr
- ☐ 3. idtr
- ☐ 4. ldtr

The correct answer is: tr

Question **12**

Correct

Mark 5.00 out of 5.00

..... is a problem wherein multiple processes are wakened only to race for a resource that can be accessed by only one of them, with the result that remaining processes must once more be put back to sleep.

Select one:

- ☐ 1. Cache Coherency
- ☒ 2. Thundering herd ✓
- ☐ 3. Race Condition
- ☐ 4. Livelock

The correct answer is: Thundering herd

Question **13**

Correct

Mark 5.00 out of 5.00

The state field of the process descriptor indicates one of the possible process states.

Select one:

- ☒ 1. 7 ✓
- ☐ 2. 8
- ☐ 3. 4
- ☐ 4. 5

The correct answer is: 7

Question **14**

Correct

Mark 10.00 out of 10.00

Exclusive processes are identified by the value in the field of the corresponding wait queue element.

Select one:

- ☐ 1. 0, func
- ☒ 2. 1, flags ✓
- ☐ 3. 0, flags
- ☐ 4. 1, func

The correct answer is: 1, flags

Question **15**

Correct

Mark 5.00 out of 5.00

The following Resource limit field in process descriptor indicates **maximum size of process address space in bytes**

Select one:

- ☒ 1. RLIMIT_AS ✓
- ☐ 2. RLIMIT_RSS
- ☐ 3. RLIMIT_DATA
- ☐ 4. RLIMIT_DATA

The correct answers are: RLIMIT_DATA, RLIMIT_AS

Question **16**

Correct

Mark 5.00 out of 5.00

Specify the default Port number associated with DNS

Answer: ✓

The correct answer is: 53

Question **17**

Correct

Mark 5.00 out of 5.00

..... state indicates whether process is either executing on a CPU or waiting to be executed.

Select one:

- ☐ 1. TASK_STOPPED
- ☐ 2. TASK_UNINTERRUPTIBLE
- ☐ 3. TASK_INTERRUPTIBLE
- ☐ 4. EXIT_ZOMBIE
- ☐ 5. EXIT_DEAD
- ☐ 6. TASK_TRACED
- ☒ 7. TASK_RUNNING ✓

The correct answer is: TASK_RUNNING

Question **18**

Correct

Mark 5.00 out of 5.00

Specify the default Port number associated with HTTP

Answer: ✓

The correct answer is: 80

Question **19**

Correct

Mark 5.00 out of 5.00

Wait queues are implemented as..... whose elements include pointers to process descriptors.

Select one:

- ☐ 1. binary search trees
- ☒ 2. doubly linked lists ✓
- ☐ 3. Merkel Patricia Tree
- ☐ 4. singly linked lists

The correct answer is: doubly linked lists

Question **20**

Correct

Mark 5.00 out of 5.00

Control flow passes from one process to another process via

Select one:

- ☒ 1. Context switching ✓
- ☐ 2. Call instruction
- ☐ 3. Jump instruction
- ☐ 4. Return instruction

The correct answer is: Context switching

Question **21**

Correct

Mark 5.00 out of 5.00

The following Resource limit field in process descriptor indicates maximum number of processes that the user can own.

Select one:

- ☐ 1. RLIMIT_MEMLOCK
- ☐ 2. RLIMIT_DATA
- ☐ 3. RLIMIT_NOFILE
- ☒ 4. RLIMIT_NPROC ✓

The correct answer is: RLIMIT_NPROC

Question **22**
Correct
Mark 10.00 out of 10.00

Identify the Layers in TCP/IP Model

Select one or more:

- ☒ 1. Internet Layer ✓
- ☐ 2. Data Link Layer
- ☒ 3. Transport Layer ✓
- ☐ 4. Presentation Layer
- ☒ 5. Application Layer ✓
- ☐ 6. Session Layer
- ☐ 7. Network Layer
- ☐ 8. Physical Layer
- ☒ 9. Network Interface Layer ✓

The correct answers are: Network Interface Layer, Application Layer, Transport Layer, Internet Layer

Question **23**
Correct
Mark 10.00 out of 10.00

The function operates on the current process: The function sets the state of the current process to TASK_UNINTERRUPTIBLE and inserts it into the specified wait queue.

Select one:

- ☒ 1. sleep_on() ✓
- ☐ 2. interruptible_sleep_on()
- ☐ 3. prepare_to_wait_exclusive()
- ☐ 4. sleep_on_timeout()

The correct answer is: sleep_on()

Question **24**
Correct
Mark 10.00 out of 10.00

The system call causes the process to block until a client connects to the server. Thus, it wakes up the process when a connection from a client has been successfully established. It returns a new file descriptor, and all communication on this connection should be done using the new file descriptor.

Answer: ✓

The correct answer is: accept()

Question **25**
Correct
Mark 5.00 out of 5.00

..... field in the process descriptor of a process P Points to the head of the list containing all children created by P.

Answer: ✓

The correct answer is: children

Question **26**

Correct

Mark 5.00 out of 5.00

..... is the final state which indicates that the process is being removed by the system because the parent process has just issued a `wait4()` or `waitpid()` system call for it.

Select one:

- ☐ 1. EXIT_ZOMBIE
- ☐ 2. TASK_TRACED
- ☐ 3. TASK_INTERRUPTIBLE
- ☒ 4. EXIT_DEAD ✓
- ☐ 5. TASK_STOPPED
- ☐ 6. TASK_RUNNING
- ☐ 7. TASK_UNINTERRUPTIBLE

The correct answer is: EXIT_DEAD

Question **27**

Correct

Mark 5.00 out of 5.00

..... state indicates that Process execution has been stopped by a debugger and process is being monitored by debugger.

Select one:

- ☐ 1. TASK_RUNNING
- ☐ 2. TASK_UNINTERRUPTIBLE
- ☐ 3. EXIT_DEAD
- ☒ 4. TASK_TRACED ✓
- ☐ 5. TASK_INTERRUPTIBLE
- ☐ 6. TASK_STOPPED
- ☐ 7. EXIT_ZOMBIE

The correct answer is: TASK_TRACED

Question **28**

Correct

Mark 10.00 out of 10.00

What is the full form of NGPT which is a package from IBM for implementing multithreading applications

Answer: ✓

The correct answer is: Next Generation Posix Threading

Question **29**

Correct

Mark 5.00 out of 5.00

The command `getconf PAGE_SIZE` returns

Select one:

- ☒ 1. 4096 ✓
- ☐ 2. 16384
- ☐ 3. 8192
- ☐ 4. 65536

The correct answer is: 4096

Question **30**

Correct

Mark 10.00 out of 10.00

For each process, Linux packs two different data structures in a single per-process memory area: a small data structure linked to the process descriptor.

What are they.....

Select one or more:

- ☒ 1. thread_info structure ✓
- ☐ 2. User Mode process stack
- ☐ 3. Task State Segment
- ☒ 4. Kernel Mode process stack ✓

The correct answers are: thread_info structure, Kernel Mode process stack

Question **31**

Correct

Mark 5.00 out of 5.00

Specify the default Port number associated with Telnet

Answer:



The correct answer is: 23

Question **32**

Correct

Mark 5.00 out of 5.00

The..... function initializes all the data structures needed by the kernel, enables interrupts, and creates another kernel thread, named process 1 (init process)

Select one:

- ☐ 1. first startup()
- ☒ 2. start_kernel() ✓
- ☐ 3. setup()
- ☐ 4. second startup()

The correct answer is: start_kernel()

Question **33**
Correct
Mark 10.00 out of 10.00

The thread_info structure is bytes long, the kernel stack can expand up to bytes.

Select one:

- ☐ 1. 92, 8100
- ☐ 2. 192, 8000
- ☒ 3. 52, 8140 ✓
- ☐ 4. 64, 8128

The correct answer is: 52, 8140

Question **34**
Incorrect
Mark 0.00 out of 10.00

For systems having 512 MB of RAM, each hash table is stored in page frames and includes entries.

Select one:

- ☐ 1. 4, 2048
- ☐ 2. 8, 4096
- ☒ 3. 2, 2048 ✗
- ☐ 4. 4, 4096

The correct answer is: 4, 2048

Question **35**
Correct
Mark 5.00 out of 5.00

..... field of the process descriptor describes what is currently happening to the process.

Select one:

- ☐ 1. run_list
- ☐ 2. thread_info
- ☒ 3. state ✓
- ☐ 4. tasks

The correct answer is: state

Question **36**
Incorrect
Mark 0.00 out of 5.00

..... field in the process descriptor of a process P have pointers to the next and previous elements in the list of the processes, those that have the same parent as P.

Answer: ✗

The correct answer is: sibling

Question **37**

Correct

Mark 5.00 out of 5.00

The flag in the CR0 allows the kernel to save and restore the FPU, MMX, and XMM registers only when really needed.

Select one:

- ☐ 1. PG
- ☐ 2. TF
- ☒ 3. TS ✓
- ☐ 4. PS

The correct answer is: TS

Question **38**

Correct

Mark 5.00 out of 5.00

The kernel thread created by process 0 executes the function, which in turn completes the initialization of the kernel.

Select one:

- ☐ 1. copy_process()
- ☒ 2. init() ✓
- ☐ 3. schedule()
- ☐ 4. do_fork()

The correct answer is: init()

Question **39**

Correct

Mark 5.00 out of 5.00

The system call used to create a light-weight process or thread is

Select one:

- ☐ 1. fork
- ☐ 2. flock
- ☐ 3. vfork
- ☒ 4. clone ✓

The correct answer is: clone

Question **40**

Correct

Mark 5.00 out of 5.00

The aim of using large number of runqueue is to allow the scheduler to select the best runnable process in constant time, independently of the number of runnable processes.

Say True or False

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **41**

Correct

Mark 5.00 out of 5.00

To speed up the search for a process descriptor corresponding to a given PID, hash tables have been introduced in Linux.

Select one:

- ☐ 1. 2048
- ☒ 2. 4 ✓
- ☐ 3. 16
- ☐ 4. 8

The correct answer is: 4

Question **42**

Correct

Mark 5.00 out of 5.00

..... system call allows both the parent and the child to share many per-process kernel data structures, such as the paging tables (and therefore the entire User Mode address space), the open file tables, and the signal dispositions.

Select one:

- ☐ 1. vfork
- ☐ 2. fork
- ☒ 3. clone ✓
- ☐ 4. execve

The correct answer is: clone

Question **43**

Correct

Mark 10.00 out of 10.00

Non-Exclusive processes are identified by the value in the field of the corresponding wait queue element.

Select one:

- ☐ 1. 1, flags
- ☐ 2. 1, func
- ☐ 3. 0, func
- ☒ 4. 0, flags ✓

The correct answer is: 0, flags

Question **44**

Correct

Mark 5.00 out of 5.00

Linux associates a different with each process or lightweight process in the system. This approach allows the maximum flexibility, because every execution context in the system can be uniquely identified.

Answer:



The correct answer is: PID

Question **45**

Correct

Mark 5.00 out of 5.00

All sleeping processes are always woken up by the kernel when the event occurs.

Select one:

- ☒ 1. Non-Exclusive ✓
- ☐ 2. Runnable
- ☐ 3. Exclusive
- ☐ 4. Ready

The correct answer is: Non-Exclusive

Question **46**

Correct

Mark 5.00 out of 5.00

..... defines how applications can create channels of communication across a network. It also manages how a message is assembled into smaller packets before they are then transmitted over the internet and reassembled in the right order at the destination address.

Select one:

- ☐ 1. HTTP
- ☒ 2. TCP ✓
- ☐ 3. IP
- ☐ 4. ARP

The correct answer is: TCP

Question **47**

Correct

Mark 10.00 out of 10.00

Indicate the correct order of the steps involved in establishing a socket on the server side:

1. Connect
2. Accept
3. Socket Creation
4. Listen
5. Send/Recv
6. Set Socket option
7. Bind

Select one:

- ☐ 1. 3, 6, 7, 2, 4, 5
- ☐ 2. 3, 6, 2, 7, 4, 5
- ☒ 3. 3, 6, 7, 4, 2, 5 ✓
- ☐ 4. 3, 7, 6, 4, 2, 5

The correct answer is: 3, 6, 7, 4, 2, 5

Question **48**

Correct

Mark 5.00 out of 5.00

Linux associates a different PID with each process or lightweight process in the system.

Say True or False

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **49**

Correct

Mark 10.00 out of 10.00

NPTL stands for

Answer: native posix thread library ✓

The correct answer is: Native POSIX Thread Library

Question **50**

Correct

Mark 5.00 out of 5.00

The following Resource limit field in process descriptor indicates maximum CPU time for the process in seconds.

Select one:

- ☐ 1. RLIMIT_DATA
- ☐ 2. RLIMIT_AS
- ☒ 3. RLIMIT_CPU ✓
- ☐ 4. RLIMIT_RSS

The correct answer is: RLIMIT_CPU

Question **51**

Correct

Mark 10.00 out of 10.00

Maximum number of PIDs that can be used in in 64-bit architectures is

Select one:

- ☐ 1. 32768
- ☐ 2. 4194303
- ☒ 3. 4194304 ✓
- ☐ 4. 65536

The correct answer is: 4194304

Question **52**

Correct

Mark 5.00 out of 5.00

..... defines how to address and route each packet to make sure it reaches the right destination.

Select one:

- ☐ 1. TCP
- ☐ 2. SNMP
- ☐ 3. ARP
- ☒ 4. IP ✓

The correct answer is: IP

Question **53**

Correct

Mark 5.00 out of 5.00

The kernel must be able to handle many processes at the same time, and process descriptors are stored in

Select one:

- ☐ 1. flash memory
- ☐ 2. kernel memory
- ☐ 3. read only memory
- ☒ 4. dynamic memory ✓

The correct answer is: dynamic memory

Question **54**

Correct

Mark 5.00 out of 5.00

A is an end point of communication between two systems on a network.

Answer: ✓

The correct answer is: Socket

Question **55**

Correct

Mark 5.00 out of 5.00

In Linux, a `task_struct` type structure is used to store

Select one:

- ☐ 1. file descriptor
- ☐ 2. segment descriptor
- ☒ 3. process descriptor ✓
- ☐ 4. page descriptor

The correct answer is: process descriptor

Question **56**

Correct

Mark 10.00 out of 10.00

In multiprocessor systems, process running on CPU 0 initializes the kernel data structures, then enables the other CPUs and creates the additional processes by means of the `copy_process()` function passing to it the value 0 as the new PID.

You need to specify only one answer which satisfies both the fill in the blanks

Answer:



The correct answer is: swapper

Question **57**

Correct

Mark 5.00 out of 5.00

The function sets up the process descriptor and any other kernel data structure required for a child's execution.

Select one:

- ☐ 1. `execve`
- ☒ 2. `copy_process` ✓
- ☐ 3. `do_fork`
- ☐ 4. `kernal_thread`

The correct answer is: `copy_process`

Question **58**

Correct

Mark 5.00 out of 5.00

Stream sockets use protocol which is a reliable, stream oriented protocol (connection-oriented)

Select one:

- ☐ 1. UDP
- ☒ 2. TCP ✓
- ☐ 3. IP
- ☐ 4. SNMP

The correct answer is: TCP

Question **59**

Correct

Mark 5.00 out of 5.00

PID of a newly created process is normally the PID of the previously created process increased by one.

Say True or False

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **60**

Correct

Mark 5.00 out of 5.00

If a child process terminates before the parent process dies then the child process is termed as

Select one:

- ☐ 1. Bad process
- ☐ 2. Dead process
- ☐ 3. Orphan process
- ☒ 4. Zombie process ✓

The correct answer is: Zombie process

Question **61**

Correct

Mark 10.00 out of 10.00

In Linux a is basically a set of lightweight processes that implement a multithreaded application and act as a whole with regards to some system calls such as `getpid()`, `kill()`, and `_exit()`

Answer: ✓

The correct answer is: thread group

Question **62**

Correct

Mark 5.00 out of 5.00

..... field in the process descriptor of a process P Points to the current parent of P (this is the process that must be signaled when the child process terminates); its value usually coincides with that of `real_parent`. It may occasionally differ, such as when another process issues a `ptrace()` system call requesting that it be allowed to monitor P

Answer: ✓

The correct answer is: parent

Question **63**

Correct

Mark 5.00 out of 5.00

In Linux, the function is used to handle the `clone()`, `fork()`, and `vfork()` system calls

Select one:

- ☐ 1. `execve`
- ☐ 2. `copy_process`
- ☐ 3. `kernal_thread`
- ☒ 4. `do_fork` ✓

The correct answer is: `do_fork`

Question **64**

Correct

Mark 10.00 out of 10.00

..... state is used to indicate that the process is suspended (sleeping) until some condition becomes true. Raising a hardware interrupt, releasing a system resource the process is waiting for, or delivering a signal are examples of conditions that might wake up the process

Select one:

- ☐ 1. TASK_TRACED
- ☐ 2. EXIT_ZOMBIE
- ☐ 3. TASK_STOPPED
- ☐ 4. TASK_RUNNING
- ☒ 5. TASK_INTERRUPTIBLE ✓
- ☐ 6. TASK_UNINTERRUPTIBLE
- ☐ 7. EXIT_DEAD

The correct answer is: TASK_INTERRUPTIBLE

Question **65**

Correct

Mark 5.00 out of 5.00

The system call used to create a light-weight process or thread is

Select one:

- ☐ 1. flock
- ☐ 2. vfork
- ☒ 3. Clone ✓
- ☐ 4. fork

The correct answer is: Clone

Question **66**

Correct

Mark 5.00 out of 5.00

Each element in the wait queue list represents a sleeping process, which is waiting for some event to occur; its descriptor address is stored in the field of wait queue element structure.

Answer: ✓

The correct answer is: task

Question **67**

Correct

Mark 5.00 out of 5.00

The following Resource limit field in process descriptor indicates **maximum number of open file descriptors**

Select one:

- ☐ 1. RLIMIT_MEMLOCK
- ☒ 2. RLIMIT_NOFILE ✓
- ☐ 3. RLIMIT_NPROC
- ☐ 4. RLIMIT_DATA

The correct answer is: RLIMIT_NOFILE

Question **68**

Correct

Mark 5.00 out of 5.00

Datagram sockets use protocol, which is unreliable and message oriented. (Connection less)

Select one:

- ☒ 1. UDP ✓
- ☐ 2. TCP
- ☐ 3. IP
- ☐ 4. ARP

The correct answer is: UDP

Question **69**

Correct

Mark 5.00 out of 5.00

..... sleeping processes are selectively woken up by the kernel when the event occurs.

Select one:

- ☐ 1. Non-Exclusive
- ☐ 2. Ready
- ☒ 3. Exclusive ✓
- ☐ 4. Runnable

The correct answer is: Exclusive

Question **70**

Correct

Mark 5.00 out of 5.00

Specify the default Port number associated with POP3

Answer:



The correct answer is: 110

Question **71**

Correct

Mark 5.00 out of 5.00

The maximum number of pages allocated for storing PID bitmap in 64-bit architectures is

Select one:

- ☐ 1. 8
- ☐ 2. 1
- ☐ 3. 64
- ☒ 4. 128 ✓

The correct answer is: 128

Question **72**

Correct

Mark 5.00 out of 5.00

If a child process terminates after the parent process dies then the child process is termed as

Select one:

- ☐ 1. Bad process
- ☐ 2. Zombie process
- ☐ 3. Dead process
- ☒ 4. Orphan process ✓

The correct answer is: Orphan process

Question **73**

Correct

Mark 5.00 out of 5.00

The size of the pidmap_array bitmap used in 32-bit architectures to indicate which PIDs are currently assigned and free is bits.

Select one:

- ☒ 1. 32768 ✓
- ☐ 2. 4096
- ☐ 3. 32767
- ☐ 4. 65536

The correct answer is: 32768

Question **74**

Correct

Mark 5.00 out of 5.00

..... command is used in Linux to list the resource limits.

Select one:

- ☐ 1. top
- ☐ 2. du
- ☐ 3. df
- ☒ 4. ulimit -a ✓

The correct answer is: ulimit -a

Question **75**

Correct

Mark 5.00 out of 5.00

The set of data that must be loaded into the registers before the process resumes its execution on the CPU is called the

Select one:

- ☐ 1. software context
- ☐ 2. context switch
- ☐ 3. task switch
- ☒ 4. hardware context ✓

The correct answer is: hardware context

Question **76**

Correct

Mark 5.00 out of 5.00

Linux Kernel uses function to perform process switch.

Select one:

- ☒ 1. schedule() ✓
- ☐ 2. kernel_thread()
- ☐ 3. do_fork()
- ☐ 4. vfork()

The correct answer is: schedule()

Question **77**

Correct

Mark 5.00 out of 5.00

..... system call creates a process that shares the memory address space of its parent. To prevent the parent from overwriting data needed by the child, the parent's execution is blocked until the child exits or executes a new program.

Select one:

- ☐ 1. fork
- ☐ 2. clone
- ☐ 3. cfork
- ☒ 4. vfork ✓

The correct answer is: vfork

Question **78**

Correct

Mark 5.00 out of 5.00

When an 80x86 CPU switches from User Mode to Kernel Mode, it fetches the address of the Kernel Mode stack from the

Answer: ✓

The correct answer is: TSS

Question **79**

Correct

Mark 5.00 out of 5.00

Specify the default Port number associated with FTP - Data

Answer: 20



The correct answer is: 20

Question **80**

Correct

Mark 5.00 out of 5.00

Specify the default Port number associated with Finger

Answer: 79



The correct answer is: 79

Question **81**

Correct

Mark 5.00 out of 5.00

The field of a wait queue element is used to specify how the processes sleeping in the wait queue should be woken up.

Select one:

- ☐ 1. task_list
- ☐ 2. flags
- ☒ 3. func ✓
- ☐ 4. task

The correct answer is: func

Question **82**

Correct

Mark 5.00 out of 5.00

The register of each CPU contains the TSSD Selector of the corresponding TSS.

Select one:

- ☐ 1. Idtr
- ☐ 2. gdtr
- ☒ 3. tr ✓
- ☐ 4. idtr

The correct answer is: tr

Question **83**

Correct

Mark 15.00 out of 15.00

Identify the layers of the OSI Model

Select one or more:

- ☒ 1. Network Layer ✓
- ☒ 2. Application Layer ✓
- ☒ 3. Presentation Layer ✓
- ☒ 4. Data Link Layer ✓
- ☐ 5. Network Interface Layer
- ☐ 6. Internet Layer
- ☒ 7. Session Layer ✓
- ☒ 8. Physical Layer ✓
- ☒ 9. Transport Layer ✓

The correct answers are: Network Layer, Transport Layer, Application Layer, Physical Layer, Presentation Layer, Data Link Layer, Session Layer

Question **84**

Correct

Mark 5.00 out of 5.00

The following Resource limit field in process descriptor indicates the maximum heap size, in bytes

Select one:

- ☐ 1. RLIMIT_FSIZE
- ☒ 2. RLIMIT_DATA ✓
- ☐ 3. RLIMIT_AS
- ☐ 4. RLIMIT_RSS

The correct answer is: RLIMIT_DATA

Question **85**

Correct

Mark 5.00 out of 5.00

..... field in the process descriptor of a process P points to the process descriptor of the process that created P or to the descriptor of process 1 (init) if the parent process no longer exists.

Answer: ✓

The correct answer is: real_parent

Question **86**

Correct

Mark 5.00 out of 5.00

In Linux PROCESS LIST is implemented as

Select one:

- ☐ 1. binary search tree
- ☐ 2. Merkel Patricia tree
- ☐ 3. singly linked list
- ☒ 4. doubly linked list ✓

The correct answer is: doubly linked list

Question **87**
Correct
Mark 10.00 out of 10.00

Maximum number of pages used for storing pidmap_array bitmap in 32-bit architectures to indicate which PIDs are currently assigned and free is.....

Select one:

- ☐ 1. 4
- ☐ 2. 8
- ☐ 3. 128
- ☒ 4. 1 ✓

The correct answer is: 1

Question **88**
Correct
Mark 5.00 out of 5.00

Process is

Select one:

- ☐ 1. a job in secondary memory
- ☐ 2. a program in High level language kept on disk
- ☒ 3. an instance of a program in execution ✓
- ☐ 4. nothing but contents of main memory

The correct answer is: an instance of a program in execution

Question **89**
Correct
Mark 5.00 out of 5.00

..... process state is used under certain specific conditions in which a process must wait until a given event occurs without being interrupted.

Select one:

- ☐ 1. EXIT_DEAD
- ☒ 2. TASK_UNINTERRUPTIBLE ✓
- ☐ 3. TASK_RUNNING
- ☐ 4. TASK_STOPPED
- ☐ 5. TASK_TRACED
- ☐ 6. TASK_INTERRUPTIBLE
- ☐ 7. EXIT_ZOMBIE

The correct answer is: TASK_UNINTERRUPTIBLE

Question **90**

Correct

Mark 5.00 out of 5.00

Version 6 of the Internet Protocol (IPv6) defines an IP address as a number.

Select one:

- ☐ 1. 64-bit
- ☒ 2. 128-bit ✓
- ☐ 3. 256-bit
- ☐ 4. 32-bit

The correct answer is: 128-bit

Question **91**

Correct

Mark 5.00 out of 5.00

In Linux, depending on the process priority, the runqueue list has been split intodifferent lists in order to make process scheduler operations more efficient.

Select one:

- ☐ 1. 6
- ☒ 2. 140 ✓
- ☐ 3. 70
- ☐ 4. 3

The correct answer is: 140

Question **92**

Correct

Mark 5.00 out of 5.00

The resource limits for the current process are stored in the current->signal->..... field, that is, in a field of the process's signal descriptor

Answer: ✓

The correct answer is: rlim

Question **93**

Correct

Mark 10.00 out of 10.00

In Linux, a part of the hardware context of a process is stored in the field of the process descriptor while the remaining part is saved in the Kernel Mode stack.

Answer: ✓

The correct answer is: thread

Question **94**

Correct

Mark 5.00 out of 5.00

A process waiting for a resource that can be granted to just one process at a time is a typical process.

Answer: exclusive



The correct answer is: exclusive

Question **95**

Correct

Mark 5.00 out of 5.00

The upper limit on the PID number in 32-bit architectures is

Select one:

- ☐ 1. 1024
- ☐ 2. 32768
- ☒ 3. 32767 ✓
- ☐ 4. 2048

The correct answer is: 32767

Question **96**

Correct

Mark 5.00 out of 5.00

Specify the default Port number associated with HTTPS

Answer: 443



The correct answer is: 443

Question **97**

Correct

Mark 5.00 out of 5.00

.....uses Copy On Write (COW) technique that allows both the parent and the child processes to read the same physical pages. Whenever either one tries to write on a physical page, the kernel copies its contents into a new physical page that is assigned to the writing process.

Select one:

- ☒ 1. fork() system call ✓
- ☐ 2. clone() system call
- ☐ 3. vfork() system call
- ☐ 4. flock() system call

The correct answer is: fork() system call

Question **98**

Correct

Mark 5.00 out of 5.00

When a User Mode process attempts to access an I/O port by means of an in or out instruction, the CPU may need to access an I/O Permission Bitmap stored in the to verify whether the process is allowed to address the port.

Answer: TSS



The correct answer is: TSS

Question **99**

Correct

Mark 5.00 out of 5.00

The function to get the PID of a calling process is

Select one:

- ☐ 1. getpgrp
- ☒ 2. getpid ✓
- ☐ 3. getppgrp
- ☐ 4. getppid

The correct answer is: getpid

Question **100**

Correct

Mark 5.00 out of 5.00

The following Resource limit field in process descriptor indicates maximum number of page frames owned by the process.

Select one:

- ☐ 1. RLIMIT_DATA
- ☒ 2. RLIMIT_RSS ✓
- ☐ 3. RLIMIT_AS
- ☐ 4. RLIMIT_MEMLOCK

The correct answer is: RLIMIT_RSS